Simulation packages for rivers, estuaries and seas

D-HYDRO Suite

Dutch Delta Systems

Installation

User Manual

D-HYDRO Suite 1D2D 1.0.0

Installation manual

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Deltares
Boussinesqweg 1
2629 HV Delft
P.O. 177
2600 MH Delft
The Netherlands

telephone: +31 88 335 82 73
fax: +31 88 335 85 82
e-mail: info@deltares.nl
www: https://www.deltares.nl

For sales contact:
telephone: +31 88 335 81 88
fax: +31 88 335 81 11
e-mail: software@deltares.nl
www: https://www.deltares.nl/software

For support contact:
telephone: +31 88 335 81 00
fax: +31 88 335 81 11
e-mail: software.support@deltares.nl
www: https://www.deltares.nl/software

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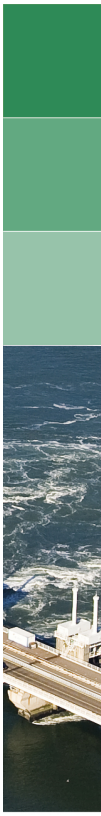


1 Installation Manual

Description:	D-HYDRO Suite 1D2D 1.0.0 (D-HYDRO 1D2D 1.0.0)
Version:	1.0.0
Status:	release, see accompanying status letter.

The D-HYDRO Suite 1D2D 1.0.0 (=D-HYDRO 1D2D 1.0.0) is the successor of D-HYDRO 1D2D 0.9.10 and earlier releases. The D-HYDRO Suite 1D2D can simulate 1D channel and sewer flow and 2D overland flow, lumped rainfall-runoff and feedback control of hydraulic structures, as well as other earlier released 2D functionality, including storm surges, hurricanes, tsunamis, detailed flows and water levels, and is capable of handling the interactions between these processes.

Please note that only the D-Flow Flexible Mesh module (1D + 2D), D-Real Time Control (1D) and D-Hydrology (0D) are currently included in the D-HYDRO 1D2D 1.0.0 Suite.

For more information, please contact software@deltares.nl.

The following distribution is available via our SOBEK Service Packages:

D-HYDRO Suite 1D2D (1.0.0):
the integrated (GUI) modeling suite based on the Delta Shell framework, with:

- the module D-Flow Flexible Mesh
- the module D-Hydrology (rainfall runoff, based on RR)
- the module D-Real Time Control (modelling feedback, based on FBC-tools)

Note: The distribution also contains the commandline/batchmode programs via the Deltares Integrated Model Runner (DIMR).
Windows installation program: <D-HYDRO Suite 1D2D (1.0.0).msi>
Linux installer package: `dhydro-1d2d-1.0.0-9_x86_64.rpm`

1.1 Installation Instructions on a Windows PC

System requirements

The minimal system requirements are specified by minimal requirements of the core components used by the system. The recommended system requirements are the specifications of the actual test environment that ensure desired usability.

Operating system		
	Required	Microsoft Windows 10 (64-bit)
	Minimal system version	1803
Processor		
	Minimal	1 GHz Intel Pentium processor or equivalent
	Recommended	Intel i7 Quad Core > 2 GHz
RAM		
	Minimal	1 GB
	Recommended	16 GB or more, depending on model sizes.
Disk space		
	D-Flow Flexible Mesh module only	500 MB (excluding model files, custom installation)
	All modules	1 GB (excluding model files, complete installation)
Screen resolution		
	Minimal	1024 x 768, 16 bit color depth
	Recommended	1920 x 1080
	Extra Remark(s)	4k Monitor is not fully supported
Software		
	Required: .NET Framework	.NET >= 4.6.1

License file and the license manager installer

The D-HYDRO software is protected by a software key (license file) dedicated to your organization. With this license file you will be able to run the D-HYDRO distribution on any of your computers, without a limitation on the number of users.

Please find the instructions on how to install the license file and license manager on our website (https://download.deltares.nl/en/lmadmin/). In the document 'License Management Installation Manual and FAQ' you will find all the information you need.

Using the installation program <.msi>

Please download the D-HYDRO Suite 1D2D (1.0.0).msi and the related documentation + tutorial data from our Download portal.

Note: For installing the D-HYDRO 1D2D 1.0.0, you need **administrator rights**.

Please double click on the downloaded installation program (<*.msi> file) to start the installation.
We recommend selecting the default options.

After the installation, you can start D-HYDRO 1D2D by clicking on the D-HYDRO 1D2D icon on the desktop or from the Windows Start-menu (*All programs* →*Deltares . . .*).

1.2 Installation Instructions on Linux using rpm

If you install the DIMR package for the first time, and intend to use parallel computing, you must install MPICH2, as follows:

```
yum install mpich
```

Assume you have the rpm file in the Downloads folder, use these steps to install the DIMR package:

```
cd ~/Downloads
rpm -ivh dhydro-1d2d-1.0.0-9_x86_64.rpm
```

Make sure to add `/opt/dhydro-1d2d_1.0.0/lnx64/bin` to your path. For example by adding this line to your `$HOME/.bash_profile`:

```
export PATH=/opt/dhydro-1d2d_1.0.0/lnx64/bin:$PATH
```

For both steps: contact your system administration if you don't have enough rights to install using yum and/or rpm. Alternatively, as a non-root user, you may extract file binary program files from the rpm and place them in a directory of your own choice, as follows:

```
cd ~/Downloads
rpm2cpio dhydro-1d2d-1.0.0-9_x86_64.rpm | cpio -idmv
export PATH=./opt/dhydro-1d2d_1.0.0/lnx64/bin:$PATH
```

Note: This version is built on a CentOS 7 (x86_64) machine. Using them on other operating systems might be problematic. Contact Deltares for alternatives (Docker, self-compilation).

1.3 Usage instructions for the commandline/batchmode programs

Note: The commandline usage is particularly useful for the beta functionality in this release: in case an example testcase does not import correctly in the Graphical User Interface, one can always directly run it from the commandline, as described below.

The commandline version of all kernels is now available via the Deltares Integrated Model Runner package (DIMR). This is automatically installed in the previous .msi step. The DIMR call in a command prompt on Windows should be:

```
set OMP_NUM_THREADS=3
"<Your installation base dir>\plugins\DeltaShell.Dimr\
kernels\x64\scripts\run_dimr.bat" dimr.xml
```

(*where the* <dimr.xml> *file for each testcase is contained in the available zip-file* <Example Testcases - D-HYDRO 1D2D 1.0.0.zip>)

On Linux the run script is called <run_dimr.sh> and command prompt is as follows:

```
set OMP_NUM_THREADS=3
<your installation basedir>/lnx64/bin/run_dimr.sh -m dimr.xml
```

Note: *Please note the extra* `-m` *option for Linux*

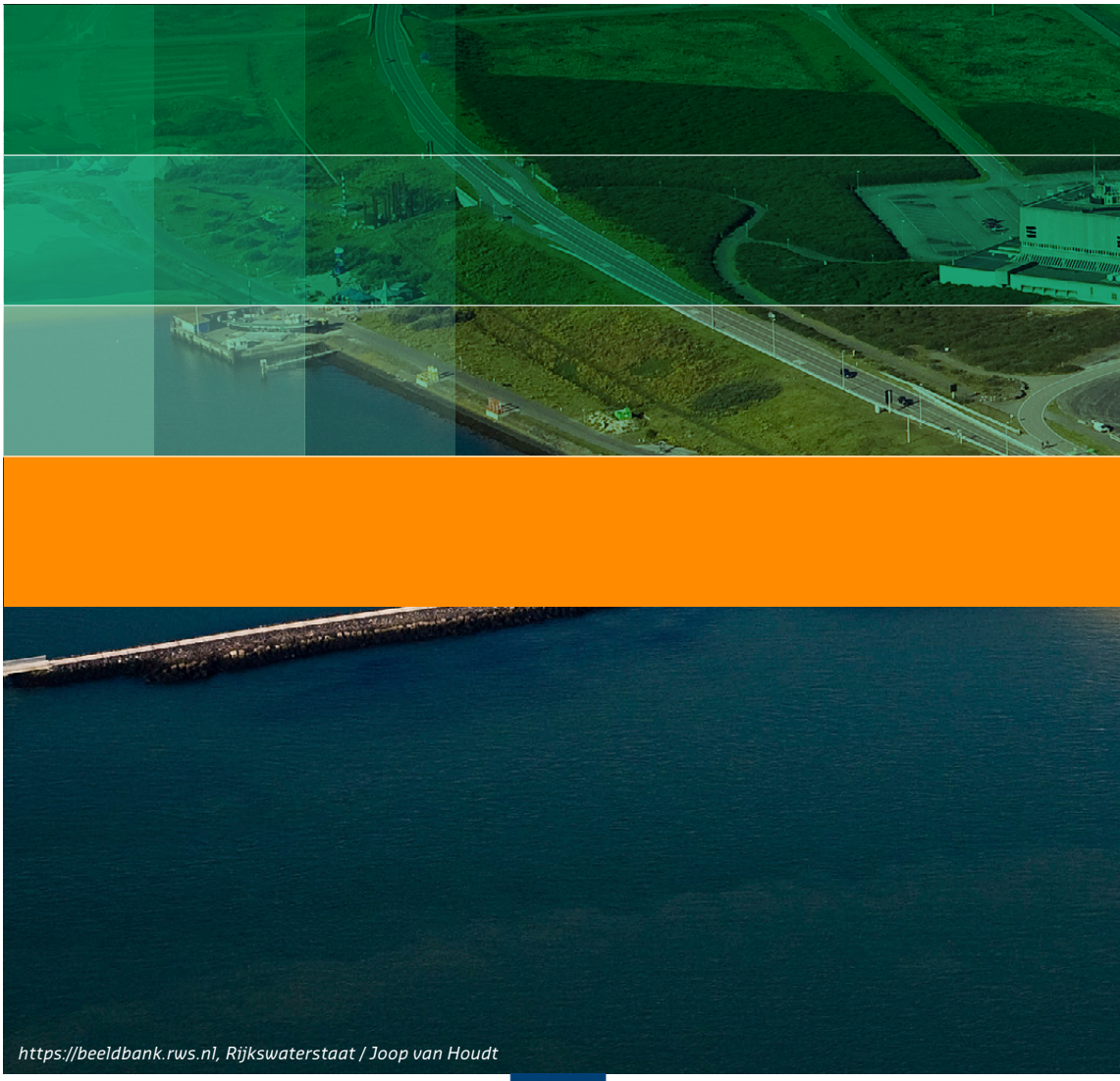
For getting started, please read the D-Flow Flexible Mesh User Manual, including step-by-step tutorials.

1.4 Source code availability

D-HYDRO is fully open source (GNU AGPL). More details can be obtained by joining our D-HYDRO beta testing – research programme.

Note: The source code for the computational core is available here:
https://svn.oss.deltares.nl/repos/delft3d/tags/delft3dfm/69614_dhydro-1d2d_rc5/

For our beta testing – research programme, please contact software@deltares.nl.



Rijkswaterstaat
Ministry of Infrastructure and Water Management

Deltares
Enabling Delta Life

PO Box 177
2600 MH Delft
Boussinesqweg 1
2629 HV Delft
The Netherlands

+31 (0)88 335 81 88